TradeMirror

EURUSD Daily Data Schema

AI Analysis Reference Document

Source: OANDA v20 REST API | Granularity: Daily (5 PM NY Close) | Pair: EUR/USD

Overview & Purpose

This document defines every column in the TradeMirror EURUSD Daily dataset. It is intended for use as a system prompt or context document when querying an AI model for trading analysis. The dataset contains 84 columns of price, volume, volatility, momentum, trend, and structural data — all calculated from OANDA live account data and updated daily at 6 PM ET.

The AI model consuming this data should use all available columns in synthesis to determine: (1) whether a LONG or SHORT position has the highest probability of a positive return, (2) optimal entry price levels, and (3) optimal exit price levels including take profit and stop loss.

Columns are organized into 14 logical groups separated by blank padding columns. Each group contributes a distinct layer of market intelligence. No single indicator should be used in isolation — the highest-probability signals emerge when multiple groups converge on the same directional bias.

AI Analysis Framework

How to Read This Dataset

Each row represents one complete trading day (5 PM NY close to 5 PM NY close, the standard Forex daily session). The most recent row is the current day. All price values are in decimal EUR/USD format (e.g. 1.0850). Pip-scaled values (True Range, ATR, Avg Delta, Avg Up, Avg Down, DM values) are multiplied by 10,000 for readability — e.g. a value of 85 means 85 pips.

Signal Synthesis Principles

When analyzing for long/short bias, weight signals as follows:

1. TREND CONFIRMATION (Groups 6, 7, 13): Is price above or below key EMAs/SMAs? Is MACD above or below signal? Is ADX rising, confirming trend strength? Is Plus DI above or below Minus DI?
2. MOMENTUM (Groups 10, 11, 12): RSI(9) and RSI(14) direction and level. StochRSI crossovers. Williams %R and CCI extremes. RSI Trend arrow direction.
3. VOLATILITY CONTEXT (Group 4): ATR defines expected daily range. Bollinger Band position shows where price is relative to its volatility envelope. Hist Vol gives broader context.
4. STRUCTURE (Groups 8, 9, 14): Pivot points define likely support/resistance. Keltner Channels define trend boundaries. Ichimoku Cloud defines the macro trend zone.
5. PRICE ACTION (Groups 1, 2): Body size, wick position, gap, and inside bar patterns provide session-level context about conviction and indecision.

Entry & Exit Guidance

ENTRY: Look for confluence of trend, momentum, and structure. A long entry is high-probability when price is above EMA(9/20/50), MACD is above signal, RSI(14) is rising but not overbought, ADX > 25, and price is near R1 having broken above it, or bouncing from a Keltner/Ichimoku support zone.

EXIT / TAKE PROFIT: Use pivot levels (R1, R2, R3 for longs; S1, S2, S3 for shorts) as primary targets. Bollinger upper/lower bands as secondary targets. Ichimoku cloud edge as a structural exit.

STOP LOSS: Place stops based on ATR(14). A 1.5x to 2x ATR stop from entry is standard. For longs, stop below the most recent swing low or below Keltner Lower. For shorts, stop above the most recent swing high or above Keltner Upper.

Column Reference by Group

Group 1 — Price (Cols A-H)

The raw OHLC data from OANDA plus derived price structure metrics. These are the foundation of all other calculations.

Col	Name	Formula / Definition	AI Trading Significance

A	Date	YYYY-MM-DD format, 5 PM NY close date	Timestamp for time-series ordering and lookback calculations
B	Open	First traded price of the session (4dp)	Gap between prev Close and Open signals overnight sentiment shift
C	High	Highest price traded during the session (4dp)	Upper boundary of the day's range; used in pivot, Ichimoku, Williams %R
D	Low	Lowest price traded during the session (4dp)	Lower boundary of the day's range; used in pivot, Ichimoku, True Range
E	Close	Last traded price of the session (4dp)	Most important price — drives RSI, MACD, EMA, SMA, and momentum calculations
F	Body	Close minus Open (4dp). Positive = bullish, negative = bearish	Magnitude and direction of body shows conviction. Large body = strong session momentum
G	Upper Wick	High minus max(Open, Close) (4dp)	Long upper wick = rejection of higher prices. Bearish signal, especially near resistance
H	Lower Wick	min(Open, Close) minus Low (4dp)	Long lower wick = rejection of lower prices. Bullish signal, especially near support

Group 2 — Session Context (Cols J-L)

Contextual metrics about how the current session opened and performed relative to the previous session.

Col	Name	Formula / Definition	AI Trading Significance
J	Gap	Open minus prev Close (4dp). Positive = gap up, negative = gap down	Forex gaps are rare but significant. A gap up with bullish follow-through is strongly bullish. Gap fills are common targets
K	Inside Bar	1 if today High <= prev High AND today Low >= prev Low, else 0	Inside bars signal consolidation and indecision. A breakout of the inside bar range is a high-probability entry trigger
L	% Change	(Close - prev Close) / prev Close x 100 (2dp)	Daily return. Used to identify momentum days. Consecutive large positive values = trend acceleration

Group 3 — Volume (Cols N-P)

OANDA tick volume data and derived volume indicators. Note: Forex volume is tick-based, not contract-based, but still provides meaningful relative comparisons.

Col	Name	Formula / Definition	AI Trading Significance
N	Volume	Raw tick volume for the session	High volume on directional days confirms the move. Low volume on trend days signals weakness
O	OBV	On Balance Volume: cumulative sum, +volume on up days, -volume on down	Rising OBV with rising price = healthy trend. OBV divergence from price = potential reversal warning
P	Vol SMA(20)	20-period simple moving average of Volume (whole number)	Compare current Volume to this average. Volume > SMA confirms significance of price move

Group 4 — Volatility (Cols R-W)

Volatility measures define the expected range of price movement. They are critical for position sizing, stop placement, and identifying breakout conditions. Note: True Range, ATR, and Avg Delta are scaled x10,000 (pip units).

Col	Name	Formula / Definition	AI Trading Significance
R	True Range	max(H-L, |H-prevC|, |L-prevC|) x 10,000 (pips, 4dp)	Single-day volatility. Large TR on a trend day confirms momentum. Large TR on a reversal day signals exhaustion
S	ATR(14)	14-period Wilder smoothed True Range x 10,000 (whole pips)	The primary tool for stop loss sizing. Stop = entry +/- 1.5-2x ATR. Also defines expected daily range for take profit targets
T	Hist Vol	20-period annualized std dev of log returns x 100 (2dp, percentage)	Macro volatility regime indicator. High HV = wider stops needed. Low HV = market may be coiling for a breakout
U	BB Upper	SMA(20) + 2 standard deviations (4dp)	Price touching or exceeding BB Upper in a trend = momentum. Price reverting from BB Upper in range = sell signal
V	BB Middle	20-period Simple Moving Average of Close (4dp)	Dynamic mean reversion target. Price tends to revert to this level. Also a support/ resistance zone
W	BB Lower	SMA(20) - 2 standard deviations (4dp)	Price touching BB Lower in a trend = momentum. Bounce from BB Lower in range = buy signal

Group 5 — Average Price & Rate of Change (Cols Y-AA)

Derived price metrics that smooth daily fluctuations and measure momentum of the average price rather than just the close.

Col	Name	Formula / Definition	AI Trading Significance
Y	Avg Price	(High + Low + Close) / 3 (4dp). Also called Typical Price	More stable than Close alone. Used as the basis for CCI and as a smoothed price reference for AI pattern detection
Z	Avg Delta	Today Avg Price minus prev Avg Price x 10,000 (pip units, 4dp)	Direction and magnitude of typical price movement. Consecutive positive deltas = uptrend in average price
AA	ROC(5)	(Close - Close[5]) / Close[5] x 100 (2dp). 5-period Rate of Change	Measures momentum over 5 days. Positive and rising = accelerating uptrend. Negative and falling = accelerating downtrend. Zero-line crossovers are signals

Group 6 — Moving Averages (Cols AC-AK)

The core trend-following indicators. Price position relative to these averages defines the macro and micro trend. EMA(12) and EMA(26) are also the building blocks of MACD in Group 7.

Col	Name	Formula / Definition	AI Trading Significance
AC	SMA(20)	20-period Simple Moving Average of Close (4dp)	Short-term trend baseline. Price above = short-term bullish. Crossovers with SMA(50) are significant signals
AD	SMA(50)	50-period Simple Moving Average of Close (4dp)	Medium-term trend. Widely watched by institutional traders. Golden cross (SMA50 > SMA200) = major bullish signal
AE	SMA(200)	200-period Simple Moving Average of Close (4dp)	The most important long-term trend indicator. Price above SMA(200) = secular uptrend. Major support/resistance
AF	EMA(9)	9-period Exponential Moving Average of Close (4dp)	Fast-reacting short-term trend. Used for precise entry timing. EMA(9) crossover of EMA(20) = short-term trend shift
AG	EMA(12)	12-period EMA of Close (4dp). Fast MACD component	Building block of MACD. Also useful as a standalone fast trend reference

AH	EMA(20)	20-period EMA of Close (4dp). Keltner Channel midline	Dynamic support/resistance in trending markets. Also the centerline of the Keltner Channel in Group 9
AI	EMA(26)	26-period EMA of Close (4dp). Slow MACD component	Building block of MACD. Slower trend baseline — crossover with EMA(12) drives MACD signal
AJ	EMA(50)	50-period EMA of Close (4dp)	Responsive medium-term trend. Faster than SMA(50). Important dynamic support in trending markets
AK	EMA(200)	200-period EMA of Close (4dp)	Long-term dynamic trend. More responsive than SMA(200). Bounce from EMA(200) in a trend is a high-probability long entry

Group 7 — MACD (Cols AM-AO)

Moving Average Convergence Divergence. Built from EMA(12) and EMA(26) in Group 6. One of the most widely used momentum and trend-following indicators.

Col	Name	Formula / Definition	AI Trading Significance
AM	MACD	EMA(12) minus EMA(26) (4dp)	Positive = short-term momentum above long-term (bullish). Negative = bearish. Direction of change is as important as sign
AN	Signal	9-period EMA of MACD line (4dp)	The trigger line. MACD crossing above Signal = bullish crossover (buy). MACD crossing below Signal = bearish crossover (sell)
AO	MACD Hist	MACD minus Signal (4dp). The histogram value	Growing positive histogram = strengthening bullish momentum. Shrinking histogram while price rises = divergence warning. Zero-line cross is the crossover signal

Group 8 — Wilder Pivot Points (Cols AQ-AV)

J. Welles Wilder's pivot point system from New Concepts in Technical Trading Systems. Calculated from the current day's OHLC. These define likely support and resistance levels for the session. R2 and S2 are the midpoints between R1/R3 and S1/S3 respectively (custom extension of Wilder's original R2/S2).

Col	Name	Formula / Definition	AI Trading Significance
AQ	R3	High + 2 x (Pivot - Low) (4dp). Strongest resistance	Major resistance. Price reaching R3 in a single session indicates exceptional bullish momentum. Often a take-profit zone for longs
AR	R2	(R1 + R3) / 2 (4dp). Intermediate resistance	Secondary resistance between R1 and R3. Useful as a partial take-profit level or a zone to watch for momentum slowdown
AS	R1	2 x Pivot - Low (4dp). First resistance	The most commonly tested resistance level. A break above R1 on volume is a bullish continuation signal. Failure at R1 is a short entry
AT	S1	2 x Pivot - High (4dp). First support	The most commonly tested support level. A bounce from S1 is a long entry. A break below S1 on volume is a bearish continuation signal
AU	S2	(S1 + S3) / 2 (4dp). Intermediate support	Secondary support between S1 and S3. Useful as a partial take-profit on shorts or a zone to watch for bounce
AV	S3	Low - 2 x (High - Pivot) (4dp). Strongest support	Major support. Price reaching S3 in a single session indicates exceptional bearish momentum. Often a take-profit zone for shorts

Group 9 — Keltner Channels (Cols AX-AZ)

ATR-based volatility envelope around EMA(20). Unlike Bollinger Bands (which use standard deviation), Keltner Channels use ATR, making them more stable in trending markets. When used together with Bollinger Bands, the Keltner/BB squeeze is a powerful breakout indicator.

Col	Name	Formula / Definition	AI Trading Significance
AX	Kelt Upper	EMA(20) + 2 x ATR(14) unscaled (4dp)	Dynamic resistance in trending markets. Price consistently above Kelt Upper = strong uptrend. Failure to reach Kelt Upper = weakening trend
AY	Kelt Mid	EMA(20) — same as col AH (4dp)	The trend centerline. Price hugging Kelt Mid from above = healthy uptrend. Cross below Kelt Mid = trend weakening
AZ	Kelt Lower	EMA(20) - 2 x ATR(14) unscaled (4dp)	Dynamic support in trending markets. BB Bands inside Keltner = squeeze/coiling. Breakout from squeeze is a high-probability directional trade

Group 10 — RSI(9) with StochRSI (Cols BB-BF)

The 9-period RSI and its Stochastic derivative. The shorter period makes RSI(9) more sensitive and faster-reacting than RSI(14). Avg Up and Avg Down are in pip units (x10,000).

Col	Name	Formula / Definition	AI Trading Significance
BB	Avg Up	Wilder smoothed average of up-day moves x 10,000 over 9 periods (1dp)	Rising Avg Up = buying pressure increasing. Use in ratio with Avg Down to understand RSI directional driver
BC	Avg Down	Wilder smoothed average of down-day moves x 10,000 over 9 periods (1dp)	Rising Avg Down = selling pressure increasing. When Avg Down >> Avg Up, RSI will be low (oversold territory)
BD	RSI(9)	100 - (100 / (1 + Avg Up / Avg Down)). Fast RSI (2dp)	Fast momentum oscillator. Above 70 = overbought. Below 30 = oversold. Divergence with price is a reversal warning. Use in conjunction with RSI(14)
BE	StochRSI %K	(RSI - LowestRSI[9]) / (HighestRSI[9] - LowestRSI[9]) x 100 (2dp)	Stochastic of RSI. Above 80 = overbought momentum. Below 20 = oversold momentum. More sensitive than RSI alone
BF	StochRSI %D	3-period SMA of StochRSI %K (2dp). Signal line	Signal line for StochRSI. %K crossing above %D from below 20 = bullish. %K crossing below %D from above 80 = bearish

Group 11 — RSI(14) with RSI Trend (Cols BH-BK)

The standard 14-period Wilder RSI — the version most used by institutional traders. The RSI Trend column provides a rule-based directional signal derived from RSI(14) overbought/ oversold levels.

Col	Name	Formula / Definition	AI Trading Significance
BH	Avg Up 14	Wilder smoothed average of up-day moves x 10,000 over 14 periods (1dp)	Building block of RSI(14). Compare to Avg Down 14 to understand the balance of buying vs selling pressure
BI	Avg Down 14	Wilder smoothed average of down-day moves x 10,000 over 14 periods (1dp)	Building block of RSI(14). Large Avg Down vs Avg Up = sustained selling pressure

BJ	RSI(14)	100 - (100 / (1 + Avg Up 14 / Avg Down 14)). Standard RSI (2dp)	The primary RSI. 70+ = overbought, 30- = oversold. Centerline (50) crossovers signal trend shifts. Look for divergence with MACD and price for reversal setups
BK	RSI Trend	Arrow indicator: up arrow or down arrow based on RSI(14) zone rules	UP arrow = RSI was overbought (>=70) or has exited oversold. DOWN arrow = RSI was oversold (<=30) or has exited overbought. Arrow persists until next zone entry. Use as a directional bias filter

Group 12 — Momentum Oscillators (Cols BM-BO)

Three complementary momentum indicators each measuring market conditions from a slightly different angle.

Col	Name	Formula / Definition	AI Trading Significance
BM	Williams %R	-100 x (HighestHigh[14] - Close) / (HighestHigh[14] - LowestLow[14]) (2dp)	Range: -100 to 0. Above -20 = overbought. Below -80 = oversold. Fast-moving — useful for timing entries within a trend identified by RSI and MACD
BN	CCI	(Typical Price - SMA20 of TP) / (0.015 x Mean Deviation) (2dp)	No fixed range. Above +100 = overbought/ strong trend. Below -100 = oversold/strong downtrend. Zero-line crossovers are trend signals. CCI divergence with price = reversal warning
BO	Momentum(10)	Close minus Close[10] (4dp, pip-scaled)	Raw 10-day price momentum. Positive and rising = accelerating uptrend. Zero-line cross = momentum shift. Compare direction with ROC(5) for multi-timeframe momentum alignment

Group 13 — Wilder Directional Movement / ADX (Cols BQ-BX)

J. Welles Wilder's complete Directional Movement System from New Concepts in Technical Trading Systems. This is the definitive trend strength and direction measurement system. DM values are in pip units (x10,000).

Col	Name	Formula / Definition	AI Trading Significance

BQ	Plus DM	Raw daily +DM x 10,000: upward movement if it exceeds downward movement (4dp)	The raw building block. Positive when bulls dominated today's movement versus yesterday
BR	Minus DM	Raw daily -DM x 10,000: downward movement if it exceeds upward movement (4dp)	The raw building block. Positive when bears dominated today's movement versus yesterday
BS	Plus DM14	14-period Wilder smoothed +DM (2dp)	Smoothed buying directional pressure over 14 days
BT	Minus DM14	14-period Wilder smoothed -DM (2dp)	Smoothed selling directional pressure over 14 days
BU	Plus DI14	Plus DM14 / ATR14 x 100 (2dp). Positive Directional Indicator	When Plus DI14 > Minus DI14 = bulls in control. The DI crossover is the primary buy/sell signal of the DMI system
BV	Minus DI14	Minus DM14 / ATR14 x 100 (2dp). Negative Directional Indicator	When Minus DI14 > Plus DI14 = bears in control. Rising Minus DI14 = increasing downward pressure
BW	DX	|Plus DI14 - Minus DI14| / (Plus DI14 + Minus DI14) x 100 (whole number)	Raw directional index before smoothing. High DX = strong directional movement. Low DX = no clear direction
BX	ADX	14-period Wilder smoothed DX (whole number)	THE trend strength indicator. Below 20 = no trend/ranging. 20-25 = trend emerging. Above 25 = trend confirmed. Above 40 = strong trend. ADX does not indicate direction — use Plus/Minus DI14 for that

Group 14 — Ichimoku Cloud (Cols BZ-CF)

The complete Ichimoku Kinko Hyo system. Originally developed for Japanese rice futures, it is now one of the most comprehensive single-indicator trend systems in Forex trading. All values are on the same row as the bar they were calculated from — displacement is a charting convention not applied here.

Col	Name	Formula / Definition	AI Trading Significance
BZ	Tenkan	(9-period High + 9-period Low) / 2 (4dp). Conversion Line	Fast signal line. Similar to a 9-period moving average of the midpoint. When price is above Tenkan = short-term bullish. Tenkan slope shows short-term momentum
CA	Kijun	(26-period High + 26-period Low) / 2 (4dp). Base Line	The key trend line. Price above Kijun = medium-term bullish. Kijun is a magnet for price. Bounce from Kijun in an uptrend = strong long entry. Flat Kijun = ranging market

CB	Senkou A	(Tenkan + Kijun) / 2 (4dp). Leading Span A (current bar value)	One edge of the Ichimoku Cloud. When Senkou A > Senkou B the cloud is bullish (green). Price above both cloud edges = strong uptrend. Cloud acts as dynamic support/resistance
CC	Senkou B	(52-period High + 52-period Low) / 2 (4dp). Leading Span B	The other cloud edge. Slower moving — represents long-term equilibrium. A thicker cloud = stronger support/resistance zone. Flat Senkou B = very strong S/R level
CD	Chikou	Current session's Close value (4dp). Lagging Span	The current close stored for cross-reference. In traditional Ichimoku, Chikou above price from 26 periods ago = bullish. Use this to assess whether the current close is historically strong
CE	Senkou A+26	Senkou A value for this bar (4dp). Represents the forward cloud edge	The projected cloud upper/lower edge. Use Senkou A+26 and Senkou B+26 together to identify the cloud zone for the current bar's date
CF	Senkou B+26	Senkou B value for this bar (4dp). Represents the forward cloud edge	When Senkou A+26 > Senkou B+26 = bullish cloud. When Senkou B+26 > Senkou A+26 = bearish cloud. The gap between them defines cloud thickness and support/resistance strength

Signal Synthesis Reference

High-Probability Long Setup Checklist

The following conditions, when aligned simultaneously, define a high-probability long entry:

- Price > EMA(9), EMA(20), EMA(50), and SMA(200) — full MA stack alignment
- MACD > Signal AND MACD Hist positive and growing
- RSI(14) between 45-65 (rising momentum, not yet overbought)
- ADX > 25 AND Plus DI14 > Minus DI14
- Price above Kijun and above Ichimoku Cloud (Senkou A+26 > Senkou B+26)
- RSI Trend arrow is UP
- Price near or just above R1 (breakout confirmation) or bouncing from S1
- Volume > Vol SMA(20) on the signal day — confirms conviction

High-Probability Short Setup Checklist

Mirror of the long checklist — all conditions reversed:

- Price < EMA(9), EMA(20), EMA(50), and SMA(200)
- MACD < Signal AND MACD Hist negative and growing more negative
- RSI(14) between 35-55 (falling momentum, not yet oversold)
- ADX > 25 AND Minus DI14 > Plus DI14
- Price below Kijun and below Ichimoku Cloud (Senkou B+26 > Senkou A+26)
- RSI Trend arrow is DOWN
- Price near or just below S1 or failing at R1
- Volume > Vol SMA(20) on the signal day

Warning Signs — Reduce Confidence

The following conditions reduce confidence in any signal and should prompt caution:

- ADX < 20 — market is ranging, trend signals are unreliable
- Inside Bar (col K = 1) — consolidation, wait for breakout
- RSI(14) divergence from price (price making new highs but RSI is not)
- MACD Hist shrinking while price continues — momentum fading
- Price between Senkou A+26 and Senkou B+26 — inside the cloud is indeterminate
- Hist Vol spiking — wider stops required, reduce position size

TradeMirror EURUSD Data Schema | Generated for AI Analysis | Data Source: OANDA v20 Live API | All prices 4 decimal places (pip precision)